

Project Screenshots

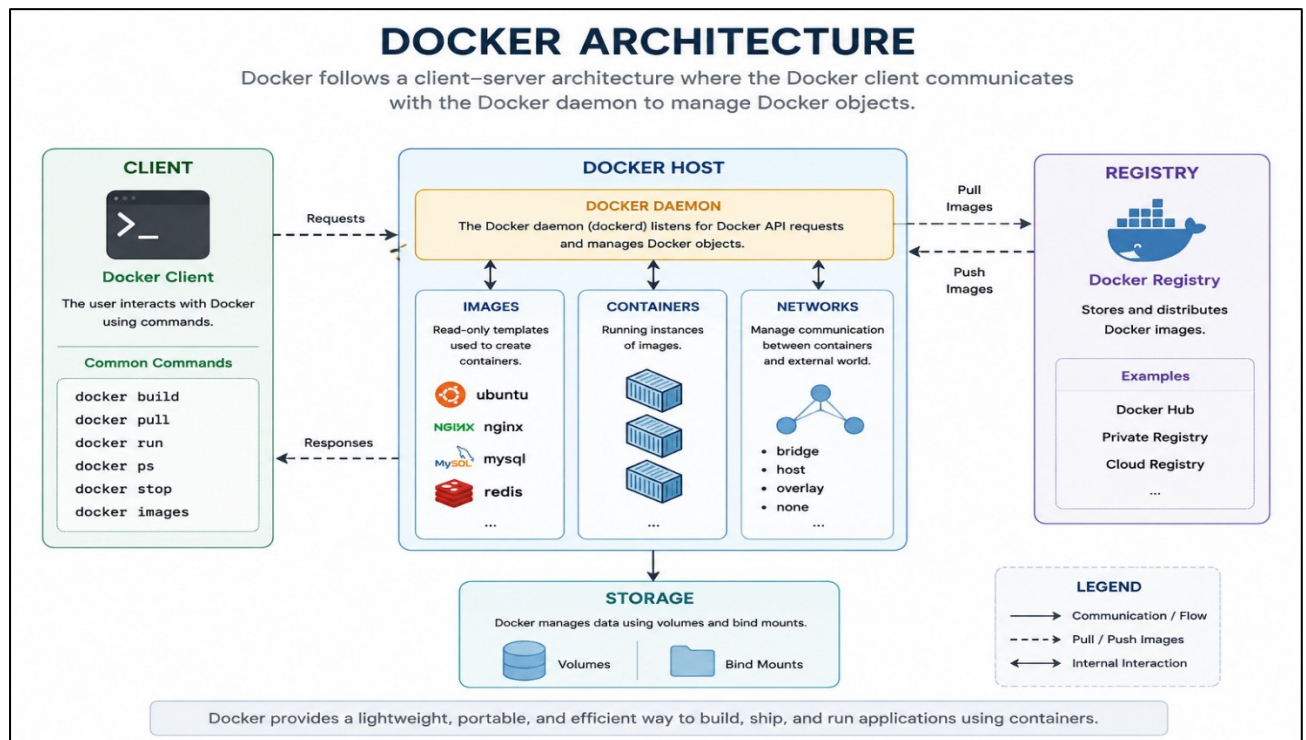


Figure: Architecture Diagram of Docker.

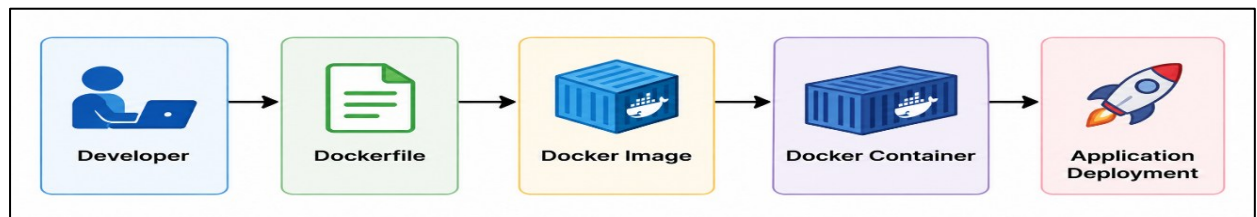


Figure: Docker Workflow

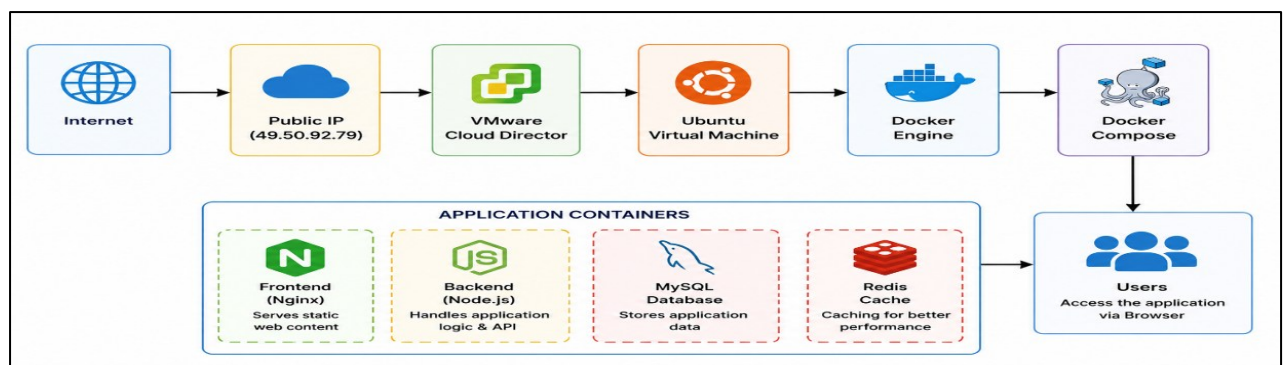


Figure: Project Architecture

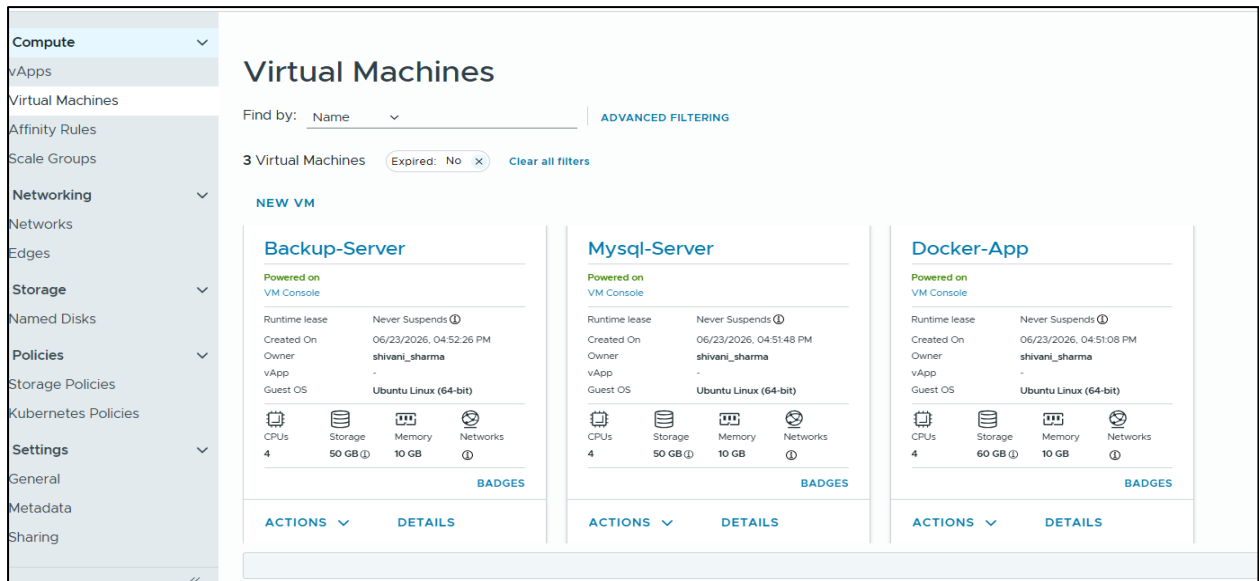


Figure: VMs Configuration

```
root@Docker-App:~# docker --version
Docker version 29.6.0, build fb59821
root@Docker-App:~# systemctl enable docker
systemctl start docker
systemctl status docker
Synchronizing state of docker.service with SysV service script with /lib/systemd
/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset:
   Active: active (running) since Tue 2026-06-23 11:48:40 UTC; 35s ago
TriggeredBy: ● docker.socket
   Docs: https://docs.docker.com
   Main PID: 54811 (dockerd)
    Tasks: 10
   Memory: 24.6M
    CPU: 610ms
   CGroup: /system.slice/docker.service
           └─54811 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/con>

Jun 23 11:48:39 docker-app dockerd[54811]: time="2026-06-23T11:48:39.672319880Z>
Jun 23 11:48:39 docker-app dockerd[54811]: time="2026-06-23T11:48:39.712057113Z>
Jun 23 11:48:39 docker-app dockerd[54811]: time="2026-06-23T11:48:39.727412971Z>
Jun 23 11:48:40 docker-app dockerd[54811]: time="2026-06-23T11:48:40.101500738Z>
Jun 23 11:48:40 docker-app dockerd[54811]: time="2026-06-23T11:48:40.110400228Z>
Jun 23 11:48:40 docker-app dockerd[54811]: time="2026-06-23T11:48:40.110584513Z>
Jun 23 11:48:40 docker-app dockerd[54811]: time="2026-06-23T11:48:40.259886762Z>
Jun 23 11:48:40 docker-app dockerd[54811]: time="2026-06-23T11:48:40.271212502Z>
```

Figure: Docker Installed.

```
root@Docker-App:~# apt install docker-compose-plugin -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
docker-compose-plugin is already the newest version (5.1.4-1~ubuntu.22.04~jammy).
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libgusb2 libmbim-glib4 libmbim-proxy libm
Use 'apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@Docker-App:~# docker compose version
Docker Compose version v5.1.4
root@Docker-App:~#
```

Figure: Docker Compose Installed.

```

root@Docker-App:~# mkdir /root/docker-project

cd /root/docker-project
root@Docker-App:~/docker-project# mkdir frontend backend

```

Figure: Project Directory Creation

```

root@Docker-App:~/docker-project# cd /root/docker-project/frontend
root@Docker-App:~/docker-project/frontend# nano index.html
root@Docker-App:~/docker-project/frontend# nano Dockerfile
root@Docker-App:~/docker-project/frontend# cat /root/docker-project/frontend/Dockerfile
FROM nginx
COPY . /usr/share/nginx/html
root@Docker-App:~/docker-project/frontend# █

```

Figure: Frontend Container Configuration

```

root@Docker-App:~/docker-project/frontend# cd /root/docker-project/backend
root@Docker-App:~/docker-project/backend# nano server.js
root@Docker-App:~/docker-project/backend# nano package.json
root@Docker-App:~/docker-project/backend# nano Dockerfile
root@Docker-App:~/docker-project/backend# cd /root/docker-project/backend

ls
Dockerfile  package.json  server.js

```

Figure: Backend Container Configuration

```

=> [backend 3/4] COPY . .
=> [backend 4/4] RUN npm install
=> [backend] exporting to image
=> => exporting layers
=> => exporting manifest sha256:75300ad6cddb8878fe61f58c7d177
=> => exporting config sha256:83fe269f51d34491ff51031ac7ea0d
=> => exporting attestation manifest sha256:85b19b3ada80ccc3
=> => exporting manifest list sha256:beb5de66b4a6a3fb487b5ee
=> => naming to docker.io/library/docker-project-backend:lat
=> => unpacking to docker.io/library/docker-project-backend:
=> [backend] resolving provenance for metadata file
[+] build 2/2
  ✓ Image docker-project-frontend Built
  ✓ Image docker-project-backend Built
root@Docker-App:~/docker-project# █

```

Figure: Image Build

```

root@Docker-App:~/docker-project# docker compose up -d
[+] up 28/28
  ✓ Image mysql:8 Pulled
  ✓ Image redis Pulled
  ✓ Network docker-project_default Created
  ✓ Volume docker-project_mysql-data Created
  ✓ Container frontend Started
  ✓ Container backend Started
  ✓ Container redis Started
  ✓ Container mysql Started
root@Docker-App:~/docker-project# █

```

Figure: Container Deployment

```
root@docker-app: ~
```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
76f249fc411e	frontend	0.00%	5.496MiB / 9.689GiB	0.06%	38.1kB / 91.8kB	0B / 24.6kB	5
b20b636259b3	redis	0.25%	6.109MiB / 9.689GiB	0.06%	3.61kB / 126B	0B / 0B	9
b769019d385f	mysql	0.44%	425MiB / 9.689GiB	4.28%	3.21kB / 126B	131MB / 566MB	34
827702dc0383	backend	0.00%	12.95MiB / 9.689GiB	0.13%	3.33kB / 126B	0B / 0B	7

```
root@Docker-App:~/docker-project# docker ps
```

CONTAINER ID	IMAGE	COMMAND
b20b636259b3	redis	"docker-entrypoint.s..."
b769019d385f	mysql:8	"docker-entrypoint.s..."
827702dc0383	docker-project-backend	"docker-entrypoint.s..."
d2f6e31c45dd	docker-project-frontend	"/docker-entrypoint...."

```
root@Docker-App:~/docker-project#
```

```
root@docker-app:~# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	In Use
docker-project-backend:latest	beb5de66b4a6	1.59GB	401MB	U
docker-project-frontend:latest	98f3f401bab4	238MB	63.1MB	U
hello-world:latest	96498ffd522e	25.9kB	9.49kB	U
mysql:8	563602a18ffd	1.12GB	254MB	U
redis:latest	0a972391db0b	208MB	56.3MB	U

```
root@docker-app:~#
```

Figure: Container Verification

```
root@docker-app:~# docker logs frontend
```

```
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/06/23 12:24:18 [notice] 1#1: using the "epoll" event method
2026/06/23 12:24:18 [notice] 1#1: nginx/1.31.2
2026/06/23 12:24:18 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/06/23 12:24:18 [notice] 1#1: OS: Linux 5.15.0-43-generic
2026/06/23 12:24:18 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
2026/06/23 12:24:18 [notice] 1#1: start worker processes
2026/06/23 12:24:18 [notice] 1#1: start worker process 29
2026/06/23 12:24:18 [notice] 1#1: start worker process 30
2026/06/23 12:24:18 [notice] 1#1: start worker process 31
2026/06/23 12:24:18 [notice] 1#1: start worker process 32
49.50.73.97 - - [23/Jun/2026:12:24:31 +0000] "GET / HTTP/1.1" 200 1779 "-" "Mozilla/5.0 (Windows
Chrome/147.0.0.0 Safari/537.36" "-"
45.198.224.186 - - [23/Jun/2026:12:43:38 +0000] "GET / HTTP/1.1" 200 1779 "-" "Mozilla/5.0" "-"
176.65.139.66 - - [23/Jun/2026:12:43:46 +0000] "GET / HTTP/1.0" 200 1779 "-" "Shodan-Pull/1.0"
151.243.11.35 - - [23/Jun/2026:12:58:51 +0000] "GET / HTTP/1.1" 400 157 "-" "curl/7.68.0" "-"
176.65.139.87 - - [23/Jun/2026:13:16:42 +0000] "GET / HTTP/1.1" 200 1779 "-" "-" "-"
```

```
^Croot@docker-app:~# docker network ls
```

NETWORK ID	NAME	DRIVER	SCOPE
476bfc854024	bridge	bridge	local
8dce10a24ca6	docker-project_default	bridge	local
a167bbdb8d42	host	host	local
caea5a008541	none	null	local

```
root@docker-app:~# docker volume ls
```

DRIVER	VOLUME NAME
local	docker-project_mysql-data

```
root@docker-app:~#
```

Figure: Docker Monitoring

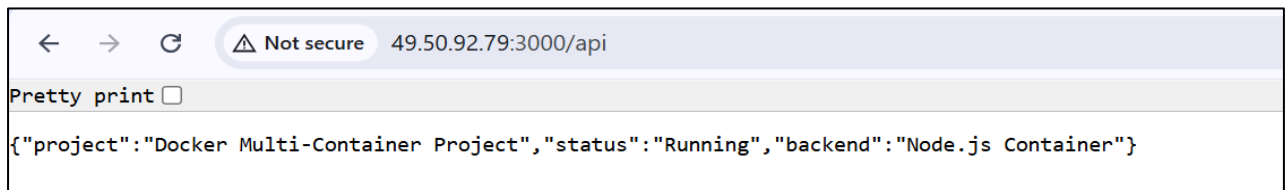
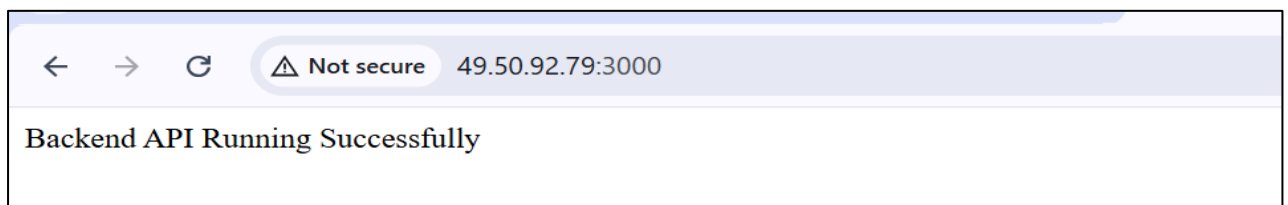


Figure: Application Access